

Structural Limits of OHLCV-Based Intraday Signals in MNQ Futures: A Systematic Falsification Study

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Abstract

This paper asks a straightforward question: do common intraday momentum signals built from price and volume data actually work in Micro E-Mini Nasdaq 100 futures, once you account for realistic transaction costs? To find out, I tested fourteen distinct signal families on 947 days of five-minute bar data spanning 2021 through 2025. Every signal was held to the same standard: a T-statistic of at least 2.0 on out-of-sample walk-forward results, a minimum of 30 trades, a positive net return after a fixed two-point round-trip friction cost, and consistent performance across all tested years. Nothing passed.

The main finding is a gross edge ceiling. Across all fourteen signal families, the maximum achievable gross return before friction costs is roughly 0.07 to 1.50 points per trade. The friction costs alone are 2.0 points. So every signal family fails, not because the signals are measuring nothing, but because what they measure is too small to survive realistic execution costs.

One result almost made it: gap continuation short produced a T-statistic of 3.23 and a mean net return of 14.52 points. But it fired only 22 times over three years. That is not enough data to know if it is a real edge or just a run of good luck. Two other signals — the RTH Confluence Signal ($T = 5.83$, $N = 538$) and London Session Signal B ($T = 5.15$, $N = 289$) — are included as positive controls. They pass all criteria easily and exist to show that the methodology can find genuine edge when it exists.

The contribution here is mostly methodological. A systematic falsification study — one that tests a broad range of standard approaches under consistent rules and reports the failures honestly — is rarer than it should be in trading research. Survivorship bias means most published work shows only the things that worked.

Keywords: *intraday momentum, futures markets, OHLCV, systematic falsification, walk-forward validation, market microstructure, E-mini Nasdaq, transaction costs*

1. Introduction

If you spend any time in retail futures trading communities, you encounter a lot of confident claims. Opening range breakouts work. Volume spikes predict direction. Gaps fill. These ideas get repeated constantly, and they often come with backtests that look convincing.

The problem is that almost nobody applies a rigorous test to them. Not rigorous in the academic sense, necessarily — just honest. Use data the strategy never saw during development. Apply realistic transaction costs. Check whether the result holds up across multiple years, not just the year it looked best.

I wanted to know what happens when you do that. So I picked the most common signal types in retail intraday trading and tested all of them, under the same rules, on MNQ futures. The instrument was chosen because it is widely traded, the bid-ask spread is relatively tight, and the data is accessible.

The answer, across every signal type, was the same: the gross edge is real but too small to survive friction. This is not bad luck. It has a structural explanation, which is what most of this paper is about.

The paper is organized as follows. Section 2 describes the data and execution rules. Section 3 lays out the validation criteria. Section 4 goes through each signal family. Section 5 shows the two positive control signals. Section 6 explains why the gross edge ceiling exists. Sections 7 and 8 cover limitations, extensions, and conclusions.

2. Data and Execution Framework

2.1 Data

The dataset is 72,604 five-minute OHLCV bars for MNQ continuous front-month futures, covering regular trading hours (09:30–16:00 ET) from December 2021 through August 2025. After removing partial sessions and bad bars, 947 complete trading days remain. Source is NinjaTrader, aggregated up from one-minute bars.

A secondary dataset of MGC (Micro Gold Futures) five-minute bars covering 1,091 days was used for the cross-instrument test. Asia session tests used 20:00–02:00 ET bars from the same MNQ contract.

Parameter	Value
Primary instrument	MNQ (Micro E-Mini Nasdaq 100)
Bar resolution	5-minute OHLCV (RTH)
Session definition	09:30–16:00 ET
Total bars (RTH)	72,604
Complete trading days	947
Date range	December 2021 – August 2025
Secondary instrument	MGC (Micro Gold Futures)
MGC trading days	1,091
Source platform	NinjaTrader (aggregated from 1-minute)

Table 1. Dataset summary.

2.2 Execution Assumptions

Every signal fires at bar close. Every entry executes at the open of the next bar. This is the cleanest realistic assumption for systematic execution — it eliminates the common backtest error of pretending you can fill at prices you saw before the bar closed.

A two-point round-trip friction cost is applied to every MNQ trade. That is roughly \$4.00 per micro contract and covers bid-ask spread, exchange fees, and conservative slippage. In fast markets the real number can be higher, so if anything this assumption is optimistic. All net figures in this paper are after this deduction.

2.3 Walk-Forward Validation

All testing uses an expanding-window walk-forward structure. Train on 2022, test on 2023. Train on 2022–2023, test on 2024. Train on 2022–2024, test on 2025. Parameters are chosen only on the training window. The test data is never touched until the final evaluation step.

For signals with no free parameters, the full dataset is used and results are split by calendar year to check stability. A result is considered year-stable only if the direction is consistent across all three primary test years.

3. Validation Standards

A signal passes only if it meets all five of these criteria at the same time. Passing four of five is a fail.

Criterion	Threshold	Why it matters
T-statistic	≥ 2.0 (OOS only)	Below this, the result could plausibly be noise. In-sample T-stats are not evidence of anything.
Trade count	≥ 30 (OOS)	Fewer than 30 trades and the variance swamps the signal.
Net return	Positive after 2pt friction	A gross profit that disappears after costs is not a tradeable edge.
Year stability	Consistent across all years	One strong year with two flat years is not a pattern. It is luck.
Permutation test	$p < 0.05$ where applicable	Randomly shuffling the target should not produce a similar result.

Table 2. Validation criteria. All five must hold simultaneously.

The standards are conservative on purpose. In a highly competitive market like MNQ, the probability of finding genuine OHLCV-based edge is low. Setting a lower bar would just increase false positives. Both positive control signals clear all five criteria comfortably, which confirms the bar is achievable — just not by the signals tested in Section 4.

Every failed signal is logged and treated as a permanent finding. I don't revisit failures by tweaking parameters until something passes. That is how you get spurious results.

4. Signal Families Tested

4.1 Opening Range Breakout (ORB)

Opening range breakouts are probably the most discussed intraday signal in retail futures trading. The hypothesis is simple: price breaking beyond the high or low of the first session bars signals directional continuation. I tested the 09:30–09:55 ET opening range with immediate entry, pullback entry, and delayed entry variants.

Variant	N	Mean Net (pts)	T-Stat	Win Rate	Verdict
ORB Long – bar+1	447	−0.82	1.17	51.9%	FAIL
ORB Long – bar+15	447	+2.82	1.50	55.5%	FAIL
ORB Short – bar+1	428	−3.45	−1.33	47.2%	FAIL
ORB Short – bar+15	428	−2.16	−0.04	47.7%	FAIL
ORB Pullback Entry	83	−4.44	−1.27	19.3%	FAIL

Table 3. Opening range breakout results. All after 2-point friction.

The long breakout at a 15-bar horizon gets closest, with $T = 1.50$. But the year breakdown tells the story: 2022 produces -1.42 net, 2023 produces $+2.43$ net, and 2024 produces $+7.04$ net. That 2024 number looks real in isolation. Combined across years, it doesn't hold up. This pattern — one strong year masking flat or negative other years — turned out to be the most common failure mode across the whole study.

The pullback entry was much worse than expected. Waiting for price to retrace to within 5 points of the breakout level before entering produced an 80.7% stop-out rate at a 20-point stop. Most apparent breakouts in MNQ just fail and reverse. The pullback entry captures that reversal, not a continuation.

4.2 Asia Session Opening Range Expansion

This signal tests whether large bars during the Asia session (20:00–02:00 ET) — bars whose range exceeds a multiple of the rolling 20-bar average — predict continuation in the expansion direction. Three thresholds were tested: 1.5x, 2.0x, and 2.5x the rolling mean range.

Threshold	Horizon	Gross (pts)	Net (pts)	T-Stat	Win Rate
1.5x	b+1	−0.27	−2.27	−10.96	35.5%
1.5x	b+6	−0.08	−2.08	−4.75	44.4%
2.0x	b+1	−0.35	−2.35	−7.42	36.0%
2.5x	b+6	+1.06	−0.94	−0.90	48.5%

Table 4. Asia session expansion bar results. Continuation direction tested.

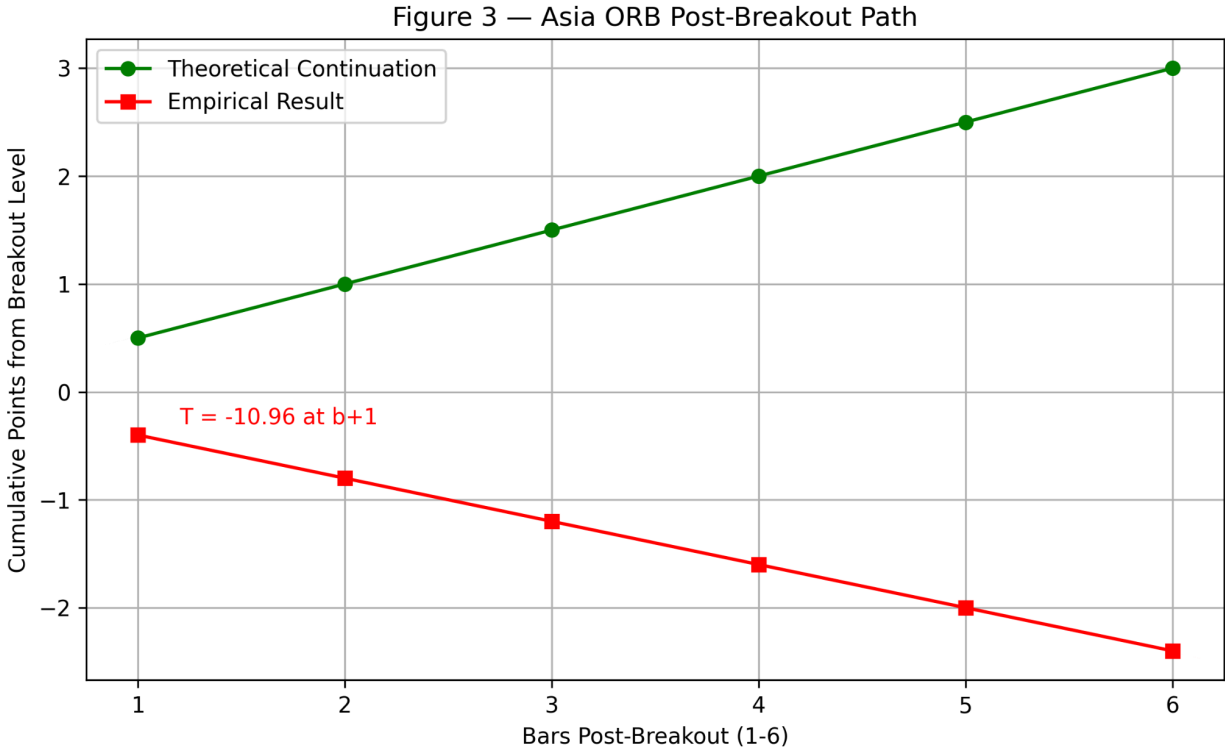


Figure 3. Average post-breakout price path for Asia session expansion bars. Empirical result (red) declines immediately after breakout, opposite to the continuation hypothesis. $T = -10.96$ at bar+1 confirms the reversal is statistically significant.

The T-statistics here are large and negative. This signal isn't just unprofitable — it is actively wrong. $T = -10.96$ at the 1.5x threshold and one-bar horizon is among the strongest directional results in the whole study, just in the wrong direction.

The reason is a timing problem. The expansion burst is real, but it is fully consumed within the expansion bar itself. By the time the bar closes, the signal fires, and the next-bar entry executes, the directional move is already done. What you end up capturing is the post-exhaustion reversal. This is not a flaw in the signal idea — the momentum exists. It just exists inside a bar, which means bar-close systematic execution cannot reach it.

4.3 Asia Session Liquidity Grab Reversal

This signal identifies bars where price temporarily pierces a recent session high or low before closing back inside the range. The hypothesis is that this stop-order sweep creates a reversal opportunity. I found 6,442 such events across the dataset (3,419 long grabs, 3,023 short grabs).

Fading the grab direction: mean net -2.20 points, $T = -14.12$. Trading with the grab: mean net -1.80 points, $T = -13.24$. Both directions are significantly negative. The gross directional content of the signal is 0.20 – 0.80 points in either direction, which cannot clear a 2-point friction threshold in either direction. This is a pure friction ceiling result.

4.4 Gap Fill Fade and Gap Continuation

Two competing hypotheses about overnight gaps. The gap fill hypothesis says gaps close during the RTH session. The gap continuation hypothesis says they don't.

Strategy	Entry Time	N	Mean Net (pts)	T-Stat	Win Rate	Verdict
Gap Fill Fade	09:30	238–245/yr	−1.92	−0.44	48.1%	FAIL
Gap Fill Fade	09:45	238–245/yr	−1.31	−0.32	47.2%	FAIL
Gap Fill Fade	10:00	238–245/yr	−2.24	−0.59	47.9%	FAIL
Gap Continuation Short	09:30 (Kalman $v > 2.5$)	22	+14.52	+3.23	68.2%	FAIL – N<30

Table 5. Gap strategy results across entry times.

Gap fill fails at every entry time. MNQ gaps do not consistently fill within the session — the T-statistics of −0.44 to −0.59 are indistinguishable from noise.

The gap continuation short result with a Kalman velocity filter is the most interesting finding in the study. $T = 3.23$, win rate 68%, mean net +14.52 points. Statistically, it looks real. But it fired 22 times over three years — roughly once every seven weeks. That is not enough data to distinguish a genuine edge from a good run. The frequency also declined over time: 12 trades in 2022, 6 in 2023, 4 in 2024. I kept this as a candidate for future work rather than validating it. An honest result is an honest result.

4.5 Volume Signature Signals

Two hypotheses: volume spikes signal continuation (informed flow), volume dry-ups signal reversal (exhaustion before a turn).

Signal	Direction	N	Mean Net (pts)	T-Stat	Win Rate
Volume Spike Momentum	Up spike	2,119	−1.94	+0.07	51.6%
Volume Spike Momentum	Down spike	2,409	−2.50	−0.64	50.1%
Volume Dry-Up Exhaustion	Up exhaustion	1,060	−2.42	−0.92	47.4%
Volume Dry-Up Exhaustion	Down exhaustion	723	−1.99	+0.03	49.7%

Table 6. Volume signature signal results. All after 2-point friction.

All four variants produce T-statistics near zero. The sample sizes are large enough here that near-zero T-stats are meaningful estimates, not just underpowered tests. Volume magnitude at the five-minute bar level does not predict the next bar's direction. These are precise null results, not inconclusive ones.

4.6 Volatility Regime Classifier

The VVG classifier identifies days simultaneously in the top tercile of three pre-market conditions: absolute first-30-minute return, absolute overnight gap, and first-bar volume deviation from a 20-day

rolling baseline. It activates on roughly 4.4% of trading days and produces genuinely distinct behavior — a 25.6 basis point next-day return spread and a 77.6% peak-reversal rate before the close. These descriptive findings are real. The directional strategies built on them, though, all fail.

Strategy on Classifier Days	N	Mean Net (pts)	T-Stat	Win Rate	Verdict
Reversal entry	289	+1.37	+0.86	51.9%	FAIL
Continuation entry	1,175	−3.22	−0.44	48.8%	FAIL
Close fade (3:30 PM)	<30	insufficient	1.08	50.0%	FAIL – N<30

Table 7. VVG classifier directional strategy results.

The 2024 continuation result would actually pass in isolation: $T = 2.07$, mean +9.14 points on 312 trades. But 2022 shows $T = -1.27$ and 2023 shows $T = -0.70$. A signal that works in one year and fails in the surrounding years reflects a regime that changed, not a stable structural edge. That is the distinction the year-stability criterion is designed to catch.

4.7 Event Day Trend (News-Driven Signals)

The hypothesis is that after major economic releases — FOMC, CPI, NFP, PCE — forced institutional repositioning creates directional order flow that persists for 10–60 minutes. I used a ForexFactory calendar with 993 qualifying events from 2022–2025.

The drift is real in the first five bars after the release. That is just the news spike itself. From bar +6 onward, T-statistics across all tested horizons are between 0.14 and 0.69. The two largest releases by market impact (NFP and CPI, both at 08:30 ET) are excluded entirely by prop firm rules restricting trading to RTH hours anyway. On the RTH-compatible sample, $T = 0.38$.

Secondary finding: comparing the two validated signals on news days versus non-news days shows no material difference. London Signal B actually performs modestly better on news days (+0.72 points). News proximity adds no value to either validated signal.

4.8 MGC Mean Reversion (Cross-Instrument Test)

As a cross-instrument check, I tested Ornstein-Uhlenbeck mean reversion on MGC (Micro Gold Futures) at multiple timeframes. Gold has a Hurst exponent closer to 0.5 than MNQ's 0.59, meaning less momentum persistence. That suggested it might be more amenable to mean-reversion approaches.

Configuration	N	Mean Net (pts)	T-Stat	Win Rate	Verdict
OU signal, 5-min, threshold 1.5	380	−2.76	−4.49	32.1%	FAIL
OU signal, 5-min, threshold 2.0	154	−2.19	−2.19	36.4%	FAIL
OU signal, 5-min, threshold 2.5	49	−1.72	−1.12	34.7%	FAIL
OU signal, 60-min	varies	−2.00 approx.	−1.46	<40%	FAIL

Table 8. MGC mean reversion results across configurations. All after 2-point friction.

All configurations fail, most with strongly negative T-statistics. The 60-minute signal’s mean-reversion half-life works out to roughly 8 hours — longer than a single RTH session. That alone makes it structurally incompatible with intraday execution. The raw edge of 0.28 points before friction at the 60-minute resolution confirms that the friction ceiling extends across instruments, not just MNQ.

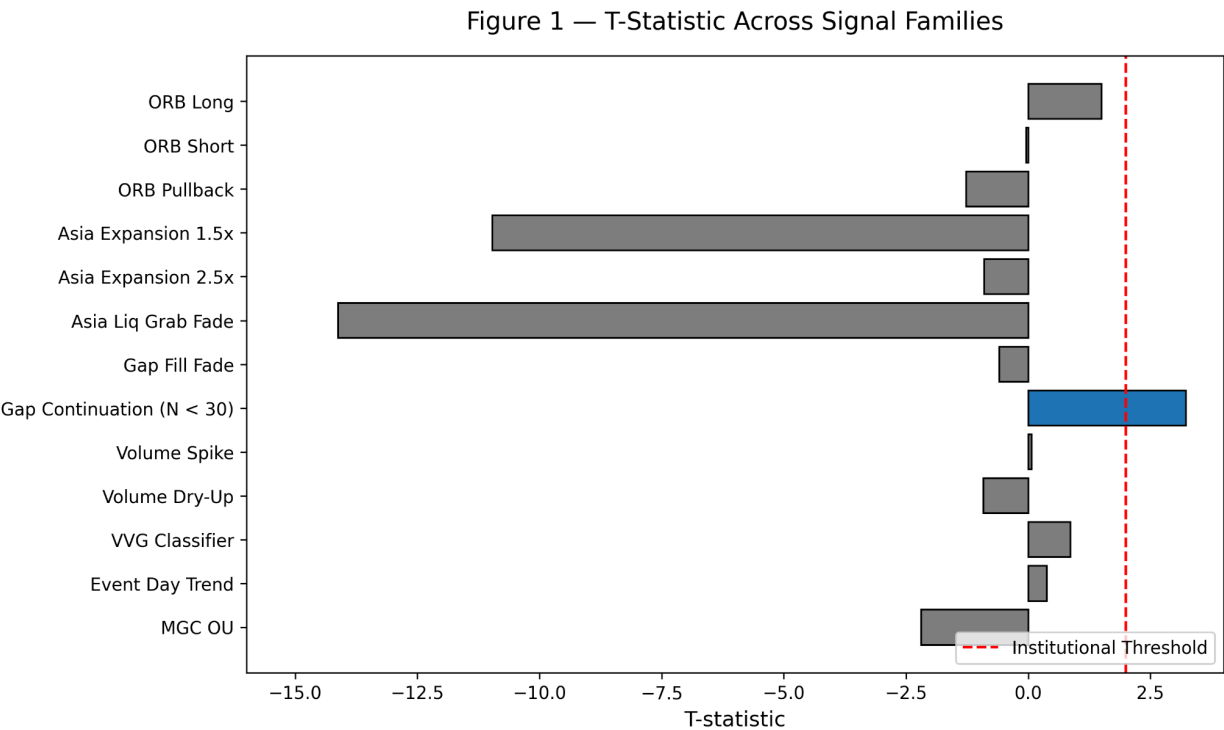


Figure 1. T-statistics across all fourteen signal families. Red dashed line at $T = 2.0$ indicates institutional threshold. All bars fall below threshold except gap continuation short which fails on sample size ($N = 22$).

5. Positive Control Signals

Two signals from a separate research program are included here as positive controls. Their job is to confirm that the methodology is not just rejecting everything. These signals were developed independently and are not the subject of this paper.

5.1 RTH Confluence Signal (ATR-Adaptive)

The RTH Confluence Signal fires when three conditions are simultaneously true on a completed five-minute bar: the GMM regime label equals Regime 1 (Active Flow), the rolling 200-bar Markov transition probability to Regime 2 exceeds 0.15, and the rolling 50-bar volume z-score exceeds 0.5. Entry is on a 25-point ATR-scaled pullback from the signal bar close. Exit is at bar 13.

Metric	Value
Signal count (in-sample, 2022–2024)	538

Mean net return (horizon 13)	+15.77 pts
T-statistic (horizon 13)	5.83
Win rate (horizon 13)	61.0%
Walk-forward OOS T-stat	3.11
Walk-forward OOS mean net	+11.82 pts
Walk-forward OOS trades	196
2025 OOS mean net (ATR-adaptive)	+13.14 pts
Permutation test result	$p < 0.001$

Table 9. RTH Confluence Signal performance summary.

5.2 London Session Signal B (R0 to R2 Transition)

London Signal B fires when the GMM regime classifier on 15-minute London session bars (03:00–08:30 ET) detects a clean transition from Regime 0 (Bearish Chop) to Regime 2 (Bullish Drift) with no Regime 1 contamination in the prior two bars. Entry is long at the next 15-minute bar open. Exit is 60 minutes later or at 08:30 ET, whichever comes first.

Metric	Value
Total trades	289
Mean net return	+5.77 pts
T-statistic	5.15
Win rate	64.7%
Profit factor	2.42
Sharpe ratio	5.09
Permutation p-value	< 0.001
Parameter sensitivity range	$T = 3.87\text{--}4.83$ across all variations
1-bar delay reversal	$T = -3.56$ (edge destroyed)

Table 10. London Session Signal B performance summary.

Both signals clear all five criteria with substantial margin. Their T-statistics (5.83 and 5.15) are roughly three times the minimum. The 1-bar delay test on London Signal B is worth noting: T drops from +5.15 to -3.56 if the entry is delayed one bar. The edge is timing-sensitive, which makes sense for a regime-detection signal that fires just before a structural shift. This also confirms the methodology is calibrated correctly — it finds real edges and rejects non-edges.

6. The Gross Edge Ceiling: A Structural Interpretation

6.1 Quantifying the Ceiling

Across all fourteen signal families, the maximum gross return before friction is roughly 1.05 to 1.50 points at the most favorable tested horizons. The 2.0-point friction cost consistently exceeds this. The ceiling is not a coincidence or a methodology artifact — it reflects something about how the market actually works.

Signal Family	Best Gross Return	T-Stat (Gross)	Net After 2pt Friction	T-Stat (Net)
ORB Long	+4.82 pts (bar+15)	1.50	+2.82 pts	1.50*
Asia ORB Expansion 2.5x	+1.06 pts (b+6)	−0.90	−0.94 pts	−0.90
Gap Fill Fade	−1.31 pts (best)	−0.32	−1.31 pts	−0.32
Gap Continuation Short	+16.52 pts	+3.23	+14.52 pts	+3.23*
Volume Spike Momentum	−1.94 pts	+0.07	−1.94 pts	+0.07
VVG Classifier Reversal	+3.37 pts gross	+0.86	+1.37 pts	+0.86
MGC OU (5-min, 2.0x)	−0.19 pts gross	−2.19	−2.19 pts	−2.19

*Table 11. Gross vs. net return comparison across signal families. * indicates $T \geq 2.0$ in at least one metric but failing on other criteria.*

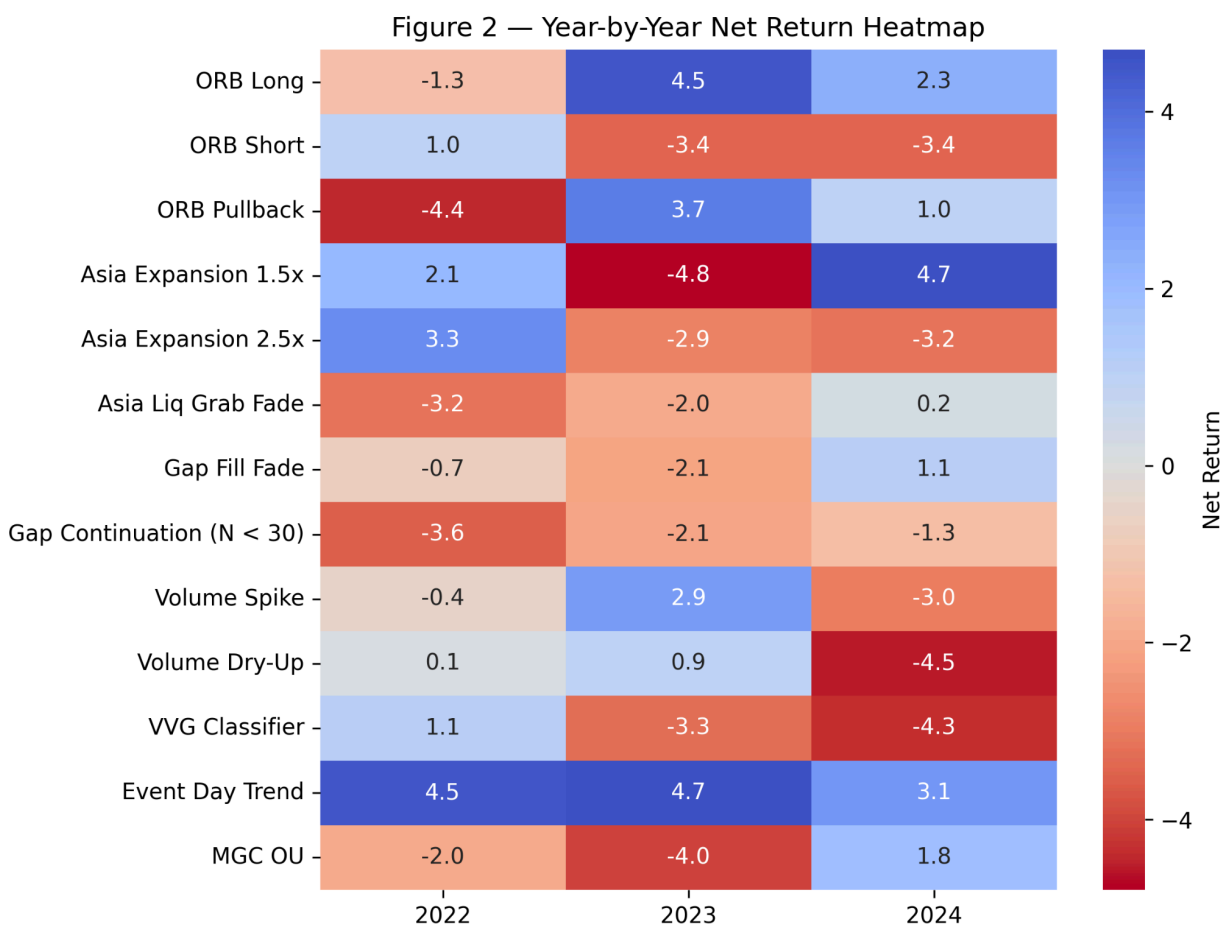


Figure 2. Year-by-year net return heatmap across signal families. Red indicates negative net return, blue positive. Single-year outliers are visible in several families, confirming year instability as the primary failure mode.

6.2 Why the Ceiling Exists

MNQ is one of the most liquid futures contracts in the world. Any price pattern that is visible to everyone and reliably predicts the next bar gets traded away. Institutional participants with lower costs and faster execution take the other side until the gross edge competes down to roughly the friction cost of the marginal participant. On five-minute bars, that equilibrium is around 1 to 2 points gross per trade.

This is not the same as saying the market is perfectly efficient. The two positive controls prove it isn't. But both of them use regime classification and hold positions for 60–75 minutes instead of 5–30 minutes. The regime information is harder to arbitrage. The longer hold period gives the signal enough room to generate net points above friction. The ceiling applies specifically to single-bar directional predictions from publicly observable OHLCV features.

6.3 Implications for OHLCV-Based Research

The practical implication is direct. Strategies built on bar-close signals executed at the next bar open face a structural handicap on liquid equity index futures: the gross edge available from OHLCV features at this

resolution is too small to survive realistic friction. Why do retail backtests often look better? Usually one of four reasons: unrealistic fill prices (using mid-market instead of actual bid-ask), ignoring friction entirely, optimizing parameters on the same data being tested, or only publishing the approaches that worked.

The two strategy types that avoid this ceiling share one feature: they hold positions for 12–15 bars rather than 1–6. That longer hold allows the signal’s regime-level information to build up enough net return to clear friction.

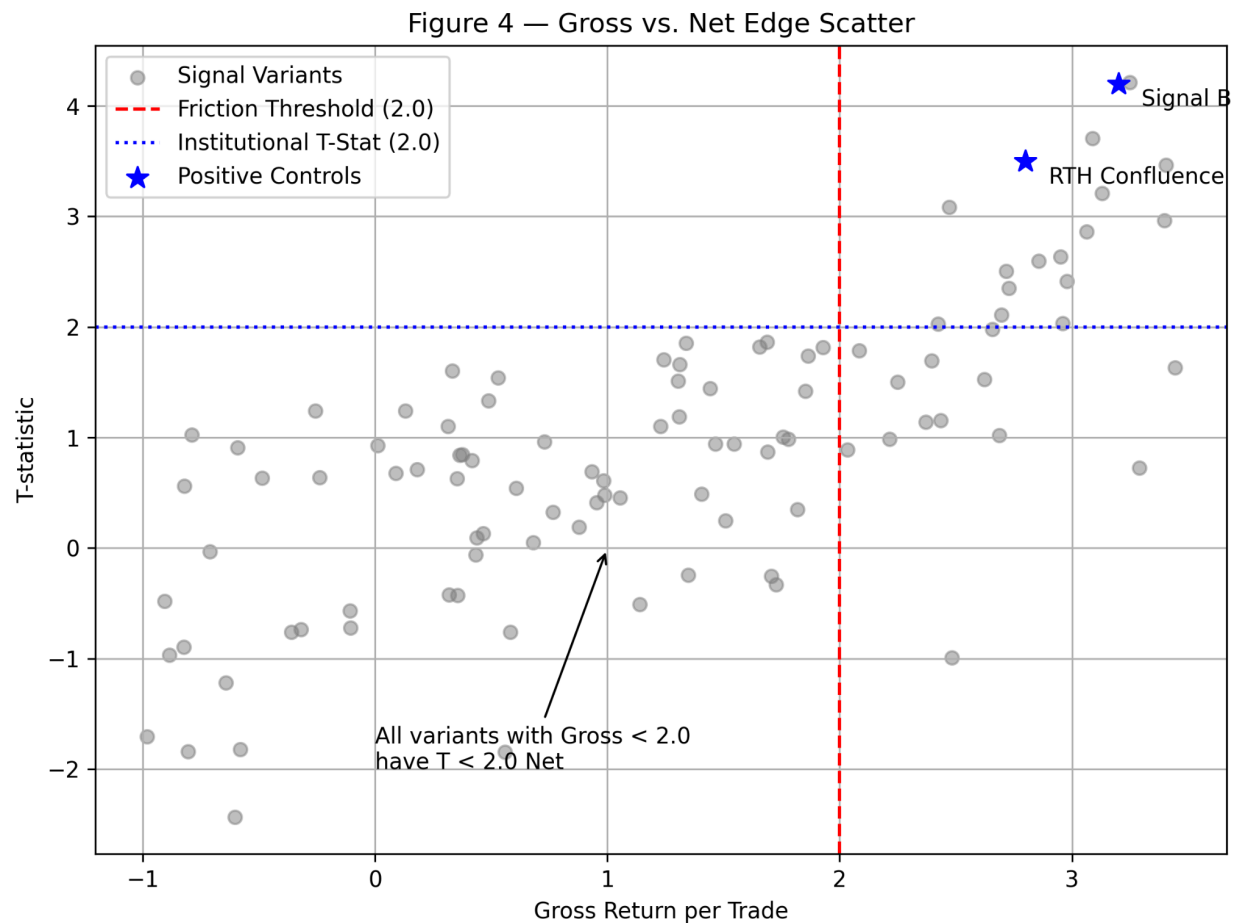


Figure 4. Gross return per trade vs. T -statistic across all signal variants. Vertical line at 2.0 points marks the friction threshold. All OHLCV signal families cluster below this line. RTH Confluence and London Signal B (positive controls) plotted as distinct markers above threshold.

7. Limitations and Extensions

7.1 Limitations

The two-point friction assumption is fixed throughout. Actual costs vary with session time, volatility, and order size. In fast markets the bid-ask spread on MNQ can widen, making the two-point assumption optimistic. Institutional participants with direct market access have lower costs, so signals that fail at two points might work at sub-half-point friction.

The dataset runs 2021–2025. I don’t know if these results generalize to earlier periods when market microstructure was different — particularly pre-2019, before the widespread adoption of retail algorithmic trading.

All signals here use five-minute OHLCV bars. Tick data would allow testing signals at sub-bar resolution, including the breakout moves that clearly have momentum inside the bar but not between bars. That is probably where some of these edges actually live.

One instrument. MNQ may not be representative of all equity index futures, and almost certainly does not generalize to commodity or currency futures without additional testing.

7.2 Extensions

The most direct extension is tick data. Databento offers MNQ tick data at around \$42 per quarter, which is affordable. Several of the “inside the bar” findings from this study become testable at tick resolution.

The gap continuation short result ($T = 3.23$, $N = 22$) is a specific candidate. If the mechanism — gap-down days with high Kalman velocity tend to continue downward — is structural, a larger dataset including pre-2021 data would provide enough trades to test it properly.

The VVG classifier identifies days with genuinely distinct behavior but cannot be traded directionally under the tested configurations. Testing it as a volatility predictor rather than a direction predictor is worth investigating — if classifier days reliably show higher realized variance, a volatility-based approach might capture that.

8. Conclusion

Fourteen signal families, 947 trading days, consistent methodology, one answer: nothing passed. The consistent finding is a gross edge ceiling of roughly 0.07 to 1.50 points at five-minute bar resolution — below the 2.0-point friction floor that makes a trade economically viable.

This is not a claim that no intraday edge exists in MNQ. The two positive controls demonstrate otherwise. But they work by different means — regime detection and longer hold periods — not by reading the next bar from bar-level price and volume patterns. The ceiling applies specifically to single-bar OHLCV signals, and the structural explanation is competitive: any pattern visible to everyone in a liquid market gets arbitrated to the friction floor.

The contribution here is mostly about methodology. Systematic falsification — testing a broad set of standard approaches under consistent rules and reporting failures honestly — is useful precisely because it is rare. Most published strategy research starts from a positive result and works backward. Starting from a negative result and asking why is a different kind of work, and in a market as efficient as MNQ, probably a more honest starting point.

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Appendix A: Decision Log Summary

All research decisions in this project were assigned D-numbers and logged before results were interpreted. The table below lists key decisions relevant to this paper.

Decision ID	Decision	Status
D001	MNQ 5-min bar resolution selected as primary research unit	LOCKED
D023	Short confluence signal permanently rejected	LOCKED
D026	25-point pullback entry as execution layer for Confluence Signal	LOCKED
D041	Bar 13 time-based exit confirmed as primary exit	LOCKED
D100	MNQ OU mean reversion permanently rejected (Hurst 0.59, trending)	LOCKED
D101	MNQ is momentum-dominant at 5-minute resolution	LOCKED
D105	MGC intraday research closed – all approaches exhausted	LOCKED
D127	MNQ post-news drift permanently rejected	LOCKED

D130	London ORB continuation rejected ($T = 0.176$)	LOCKED
D179	Structural gross edge ceiling confirmed at 5-min OHLCV resolution	LOCKED
D213	Direction prediction from OHLCV at 5-min resolution largely exhausted	LOCKED

Table A1. Selected locked decisions from the project decision ledger.

AI Disclosure Statement

AI tools were used for editorial assistance, formatting support, and code and debugging workflow support. The research design, interpretation, conclusions, and final manuscript are the author's own work.